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NSG 503 — Past Questions & Materials

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Institution: Osun State University, Osogbo

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Paper 1 of 5 — NSG 503 - NEONATAL ASPHYXIA

Level 500 • First Semester • Handout

OSUN STATE UNIVERSITY
COLLEGE OF HEALTH SCIENCES
FAULTY OF NURSING SCIENCE

COURSE CODE : NSG 503

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INTRODUCTION

Neonatal asphyxia, also known as perinatal asphyxia, is a condition that occurs when a newborn baby is

deprived of oxygen before, during, or after birth. This can lead to a range of health problems, from mild

to severe.

Causes of Neonatal Asphyxia :

1. Prolonged labor: Prolonged labor can cause fetal distress, leading to asphyxia.
2. Umbilical cord complications: Umbilical cord prolapse, cord compression, or cord rupture can reduce oxygen supply to the fetus.
3. Placental abruption: Separation of the placenta from the uterus can reduce oxygen supply to the fetus.
4. Maternal medical conditions: Conditions like hypertension, diabetes, or cardiac disease can increase the risk of neonatal asphyxia.
5. Birth trauma: Trauma during delivery, such as a difficult delivery or forceps use, can cause asphyxia.

Signs and Symptoms of Neonatal Asphyxia

1. Blue discoloration of the skin (cyanosis): Lack of oxygen can cause the skin to turn blue.
2. Slow or absent breathing: Asphyxiated newborns may have slow or absent breathing.
3. Low heart rate: Asphyxia can cause a slow heart rate.
4. Limpness or stiffness: Newborns with asphyxia may be limp or stiff.
5. Seizures: In severe cases, asphyxia can cause seizures.

Consequences of Neonatal Asphyxia

1. Brain damage: Lack of oxygen can cause brain damage, leading to developmental delays or disabilities.
2. Respiratory problems: Asphyxia can cause respiratory problems, such as chronic lung disease.
3. Cardiovascular problems: Asphyxia can increase the risk of cardiovascular problems, such as heart failure.
4. Death: In severe cases, neonatal asphyxia can be fatal.

Prevention and Treatment of Neonatal Asphyxia

1. Proper prenatal care: Regular prenatal check-ups can help identify potential risks.
2. Monitoring during labor: Continuous monitoring during labor can help detect fetal distress.
3. Prompt delivery: Prompt delivery can help prevent asphyxia.
4. Resuscitation: Resuscitation techniques, such as CPR and oxygen therapy, can help restore oxygen supply.
5. Cooling therapy: Cooling therapy can help reduce brain damage.

Resuscitation Techniques for Neonatal Asphyxia

1. Initial Assessment: Quickly assess the newborn's airway, breathing, and circulation (ABCs).
2. Clearing the Airway: Clear the airway by suctioning or using a bulb syringe to remove meconium or other debris.
3. Ventilation: Provide ventilation using a bag-valve-mask (BVM) device or a T-piece resuscitator.
4. Chest Compressions: Perform chest compressions if the heart rate remains below 60 beats per minute despite adequate ventilation.
5. Medications: Administer medications such as epinephrine, sodium bicarbonate, or naloxone as indicated.

Long-Term Effects of Neonatal Asphyxia

1. Cerebral Palsy: Neonatal asphyxia is a leading cause of cerebral palsy.
2. Developmental Delays: Asphyxiated newborns may experience developmental delays, including cognitive, motor, and language delays.
3. Seizures and Epilepsy: Neonatal asphyxia can cause seizures and increase the risk of developing

epilepsy.

4. Visual and Hearing Impairments: Asphyxiated newborns may experience visual and hearing impairments.

5. Behavioral and Emotional Problems: Neonatal asphyxia can increase the risk of behavioral and emotional problems, such as attention deficit hyperactivity disorder (ADHD) and anxiety disorders.

6. Growth and Nutrition Problems: Asphyxiated newborns may experience growth and nutrition problems, including failure to thrive.

7. Increased Risk of Infections: Neonatal asphyxia can increase the risk of infections, particularly respiratory and gastrointestinal infections.

Neonatal asphyxia can be classified based on various factors, including the severity of the condition, the timing of the asphyxial event, and the underlying cause. Here are some common classification systems:

Classification Based on Severity

1. Mild Asphyxia: Apgar score 4-6, minimal respiratory distress, and no significant neurological impairment.

2. Moderate Asphyxia: Apgar score 1-3, significant respiratory distress, and some neurological impairment.

3. Severe Asphyxia: Apgar score 0, severe respiratory distress, and significant neurological impairment.

Classification Based on Timing

1. Antepartum Asphyxia: Asphyxia occurring before birth, often due to maternal medical conditions or placental insufficiency.

2. Intrapartum Asphyxia: Asphyxia occurring during labor and delivery, often due to umbilical cord compression or fetal distress.

3. Postpartum Asphyxia: Asphyxia occurring after birth, often due to respiratory distress or cardiac problems.

Classification Based on Underlying Cause

1. Hypoxic-Ischemic Encephalopathy (HIE): Asphyxia caused by lack of oxygen and blood flow to the brain.

2. Respiratory Distress Syndrome (RDS): Asphyxia caused by surfactant deficiency and respiratory distress.

3. Cardiovascular-Related Asphyxia: Asphyxia caused by cardiac problems, such as congenital heart defects.

4. Infection-Related Asphyxia: Asphyxia caused by infections, such as pneumonia or sepsis.

Classification Based on Apgar Score

1. Apgar Score 0-3: Severe asphyxia, often requiring immediate resuscitation.
2. Apgar Score 4-6: Moderate asphyxia, often requiring oxygen therapy and close monitoring.
3. Apgar Score 7-10: Mild asphyxia, often requiring minimal intervention.

Note: The Apgar score is a widely used assessment tool to evaluate the newborn's physical condition

immediately after birth.

Paper 2 of 5 — ADVANCED MATERNAL AND CHILD HEALTH

COURSE CODE: NSG 503

Level 500 • First Semester • Handout

OSUN STATE UNIVERSITY, OSOGBO

COLLEGE OF HEALTH SCIENCES

FACULTY OF MATERNAL AND CHILD HEALTH

DEPARTMENT OF MATERNAL AND CHILD HEALTH NURSING

TOPIC: PRE, POST-TERM, AND LIGHT FOR DATE INFANTS; TUBE FEEDING

COURSE TITLE: ADVANCED MATERNAL AND CHILD HEALTH

COURSE CODE: NSG 503

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INTRODUCTION

- Tube feeding (also called gavage feeding) is the process of delivering nutrition (breast milk or formula) directly into a baby's stomach through a nasogastric (NG) or orogastric (OG) tube when oral feeding is not possible. It's a way to provide nutrition to infants who are unable to eat by mouth. Oral medicines can also be given through the tube... It's often used for babies such as premature babies, sick babies, babies who have difficulty sucking or swallowing, and babies who need extra nutrition for growth.

INDICATIONS

- Prematurity

Premature babies may not have the coordination to suck, swallow, and breathe effectively until they are older.

- Illness

Sick babies may not have enough energy or nutrients to breastfeed or bottle feed.

- Swallowing difficulties

Babies may have difficulty swallowing due to structural abnormalities or other conditions.

- Nutritional needs

Babies may have increased nutritional needs due to chronic illness or other conditions.

Other conditions that may require tube feeding includes: severe gastroesophageal reflux, Failure to thrive, Neurologic and neuromuscular impairment, and Anatomical congenital

Types of Tube Feeding in Infants

1. Orogastric tube (OGT): A tube inserted through the mouth and guided into the stomach.
2. Nasogastric tube (NGT): A tube inserted through the nose and guided into the stomach.

3. Gastrostomy tube (G-tube): A tube inserted directly into the stomach through a surgical

- Feeding Methods

1. Bolus Feeding (Intermittent Feeds)

Given every 2-3 hours through a syringe, mimicking normal feeding patterns.

2. Continuous Feeding

Using an enteral feeding pump for slow, steady delivery in infants who cannot tolerate large volumes at once.

Tube Feeding in Preterm Infants

- Preterm is an infant born at or before 37 weeks of pregnancy or (less than 259 days) of gestational age as opposed to full term pregnancy of 40 weeks.

Benefits of Tube Feeding in Preterm Infants

- 1. Nutritional support: Tube feeding ensures preterm infants receive adequate nutrients, vitamins, and minerals for growth and development.
- 2. Weight gain: Tube feeding helps preterm infants gain weight and achieve optimal growth rates.
- 3. Reduced risk of malnutrition: Tube feeding reduces the risk of malnutrition and related complications, such as infections and developmental delays.
- 4. Supports gut maturation: Tube feeding can help promote gut maturation and reduce the risk of necrotizing enterocolitis (NEC).

Special Considerations for Tube Feeding in Preterm Infants

1. Tube Selection: Choose orogastric tubes for very preterm infants to reduce nasal trauma.
2. Feeding Regimen: Use small, frequent feedings with gradual volume advancement.
3. Nutrition Type: Prefer expressed breast milk, supplemented with fortifiers as needed.
4. Feeding Method: Decide between bolus and continuous feeding based on tolerance.
5. Individualization: Tailor the feeding plan based on the infant's clinical status with a multidisciplinary approach.

Tube Feeding in Post-term Infants

A postterm newborn is a baby delivered at or after 42 weeks of gestation (294 days). A pregnancy that is between 41 and 42 weeks is called late-term.. Although post-term infants (born after 42 weeks gestation) are usually well-developed, some may require tube feeding due to birth complications or health issues.

Benefits of Tube Feeding in Post-Term Infants

1. Supports Nutrition – Ensures adequate intake when feeding is impaired due to birth complications.
2. Prevents Hypoglycemia – Provides steady glucose supply, especially in large-for-gestational-

age (LGA) infants.

3. Aids Recovery from Meconium Aspiration Syndrome (MAS) – Prevents aspiration risk in infants needing respiratory support.

4. Supports Neurological Impairment – Helps babies with birth trauma, seizures, or hypotonia receive proper nutrition.

Special Considerations for Tube Feeding in

Post-Term Infants

1. Feeding Ability: Assess the infant's oral feeding readiness to determine if tube feeding remains necessary.

2. Tube Choice: Use the least invasive tube option appropriate for the infant's condition.

3. Volume and Rate: Administer feeding volumes and rates tailored to the infant's tolerance.

4. Family Education: Instruct caregivers on proper tube care and transitioning strategies.

5. Individualized Plan: Customize the feeding regimen based on the infant's specific clinical and nutritional needs.

Light Weight for Date Infant

- A light-for-date infant (also known as small for gestational age, SGA) is a baby whose birth weight is less than the 10th percentile for their gestational age. They are born at term (37-42) weeks but weigh less than the normal range for their age which is often 2.5-3.5kg

- Benefits

- 1. Ensures Adequate Growth – Provides essential nutrients for weight gain and development.

- 2. Prevents Hypoglycemia – Maintains stable blood sugar levels due to low fat and glycogen stores.

Special Considerations for Tube Feeding in

Light for Dates Infants

- Nutritional Needs: Adjust feed density to promote catch-up growth.

- Feeding Tolerance: Monitor for signs of intolerance or gastrointestinal distress.

- Tube Size: Use a smaller, gentle tube to reduce mucosal trauma.

- Gradual Advancement: Increase feed volumes slowly to allow gut adaptation.

- Growth Monitoring: Regularly track weight and other growth parameters.

- Parental Guidance: Educate caregivers on proper tube care and monitoring signs.

- Team Coordination: Collaborate with specialists for an individualized

feeding plan.

Procedure for Tube Feeding

1. Preparation

- Wash hands and gather supplies (sterile feeding tube, syringe, formula/breast milk, pH paper).
- Ensure correct fluid volume as per neonatologist's prescription.

2. Insertion of Tube

- Position the baby in a semi-upright position.
- Insert the tube gently through the nostril or mouth into the stomach.
- Confirm placement by:
- Aspirating stomach contents and checking pH (≤ 5.5 confirms gastric

PROCEDURE(CONTD)

3. Administering Feed

Use gravity feeding or a syringe push method.

Feed slowly over 10–15 minutes to prevent regurgitation.

Monitor baby's response for signs of discomfort or aspiration.

4. Post-Feeding Care

Keep baby in an upright position for 20–30 minutes to prevent reflux.

Burp the baby gently if tolerated.

Flush tube with sterile water to prevent clogging.

5 Monitoring -Blood Sugar Levels: Monitor for hypoglycemia (low blood sugar).

- Signs of Aspiration: Look for coughing, choking, or respiratory distress.

Risks of Tube Feeding

- Irritation of the nose, mouth, or stomach
- Stuffy nose or infection of the nose
- Tube may move out of the stomach
- Tube may puncture the esophageal tissue or enter the lungs

- Aspiration Pneumonia
- Abdominal Distension
- Nasal or Oral Mucosa Irritation

Tube feeding care

- Check the position of the tube before feeding
- Tape the tube to the nose or mouth to keep it in place
- Flush the tube with warm water to prevent clogging
- Keep the infant upright or inclined to prevent regurgitation

General Midwifery Roles for Preterm, Post-

term, and Light-for-Date Infants

1. Assessment and Preparation

- Evaluate the infant's gestational age, birth weight, and feeding readiness.
- Assess sucking, swallowing, and breathing coordination to determine feeding capability.
- Check for signs of feeding intolerance such as abdominal distension, vomiting, or gastric residuals.
- Confirm proper tube placement via aspiration test and auscultation before each feed.

2. Feeding Administration

- Use a nasogastric (NG) or orogastric (OG) tube for feeding.
- Preferably administer expressed breast milk (EBM) due to its immune and nutritional benefits.

General Midwifery Roles for Preterm, Post-term, and Light-for-Date Infants(CONTD)

Feed in small, frequent volumes to prevent overfeeding, aspiration, and metabolic stress.

- Employ gravity feeding or slow bolus feeding via syringe to regulate flow and prevent complications.

3. Monitoring and Complication Prevention

- Regularly check for signs of feeding intolerance (vomiting, gastric residuals, distension).
- Observe for symptoms of hypoglycemia (jitteriness, weak cry, apnea, cyanosis).
- Monitor for necrotizing enterocolitis (NEC) by assessing stool changes, abdominal girth, and temperature instability.
- KFeed in small, frequent volumes to prevent overfeeding, aspiration, and metabolic stress.
- Employ gravity feeding or slow bolus feeding via syringe to regulate flow and prevent complications.

General Midwifery Roles for Preterm, Post-term, and Light-for-Date Infants(CONTD)

4. Parental Support and Education

- Educate mothers on expressing and storing breast milk to ensure continued nutrition.
- Encourage kangaroo mother care (KMC) to enhance bonding, milk production, and thermoregulation.
- Guide parents on safe feeding transitions, from tube feeding to direct breastfeeding or bottle feeding.
- Provide emotional support, as parents of high-risk neonates may experience anxiety and stress.

Conclusion

Tube feeding is essential for preterm, post-term infants, and light for date infants who cannot feed orally due to birth complications. Tube feeding plays a critical role

in ensuring adequate nutrition for preterm, post-term, and light-for-date infants who are unable to feed orally. It supports weight gain, prevents malnutrition, and reduces complications associated with feeding difficulties. Proper monitoring, correct tube placement, and gradual weaning are essential to ensure safe and effective feeding. With proper medical care and nutritional support, these infants can develop and thrive until they are ready for full oral feeding. Proper technique and monitoring help prevent complications and support growth. Also, gradual weaning to oral feeding ensures a smooth transition to normal feeding and nurses, to ensure optimal tube feeding management.

Paper 3 of 5 — ADVANCED MATERNAL AND CHILD HEALTH NURSING COURSE CODE: NSG 503

Level 500 • First Semester • Handout

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COLLEGE OF HEALTH SCIENCES

FACULTY OF NURSING SCIENCE

DEPARTMENT OF MATERNAL AND CHILD HEALTH NURSING

TOPIC: CHROMOSOMAL ABNORMALITIES- TRISOMY 13, TRISOMY 18, TRISOMY 21 AND PHOCOMELIA

COURSE TITLE: ADVANCED MATERNAL AND CHILD HEALTH NURSING

COURSE CODE: NSG 503

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CHROMOSOMAL ABNORMALITIES

Introduction

Chromosomes are thread-like structures made of DNA and proteins, found in the nucleus of cells. They hold the genetic instructions needed for growth, development, and overall function. Humans usually have 46 chromosomes, arranged in 23 pairs—one set from each parent. Chromosomal abnormalities occur when the number or structure of chromosomes deviates from normal. These changes can happen during the formation of reproductive cells (sperm or eggs), at fertilization, or in early stages of embryo development

Types of Chromosomal Abnormalities

1. Numerical Abnormalities: Changes in the number of chromosomes.

- Trisomy: An extra chromosome is present (e.g., Down syndrome, or Trisomy 21).
- Monosomy: A chromosome is missing (e.g., Turner syndrome, or monosomy X).

2. Structural Abnormalities: Changes in the shape or structure of chromosomes.

- Deletions: A piece of the chromosome is missing e.g Cri-du-chat syndrome (deletion on chromosome 5).
- Duplications: An extra copy of a chromosome segment.
- Inversions: A section of the chromosome is reversed.
- Translocations: Sections of chromosomes are swapped between non-matching chromosomes e.g Chronic myelogenous leukemia (caused by a translocation).

Causes of Chromosomal Abnormalities

1. Errors During Cell Division
2. Nondisjunction: Chromosomes fail to separate properly.
3. Errors during crossing over (when chromosomes exchange DNA).
4. Environmental Factors: Radiation, chemicals, or infections during pregnancy.
5. Parental Age: Advanced maternal or paternal age increases the likelihood of abnormalities.

Diagnosis and Detection

Prenatal Tests:

- Amniocentesis
- Chorionic Villus Sampling (CVS)
- Non-Invasive Prenatal Testing (NIPT)

Postnatal Tests:

- Karyotyping.
- Chromosomal Microarray Analysis

TRISOMY 13

Trisomy 13 (Patau syndrome) is a rare genetic condition that occurs when the 13th chromosome appears three times (trisomy) instead of two times in a person's DNA. Trisomy 13 affects the development of the face, brain and heart, along with physical growth abnormalities throughout a child's body. Symptoms of trisomy 13 can be life-threatening and the condition poses a risk of miscarriage or death before the child turns 1 year old.

Clinical Features of Trisomy 13

It affects several different parts of the body and range in severity for each person diagnosed with the condition. They include:

- Heart abnormalities present at birth (congenital).
- Physical growth irregularities with many cases targeting the spinal cord.
- Severe issues with cognitive function.
- Underdeveloped internal organs.

Symptoms can be life-threatening, nearly 80% of babies diagnosed with Trisomy 13 don't survive past their first year. Those that do survive may face more life-threatening complications after their first year including an increased risk of developing cancer and seizures

Physical signs

Physical signs of trisomy 13 include:

- i. Cleft lip or cleft palate.
- ii. Difficulty gaining weight.
- iii. Extra fingers or toes (polydactyly).
- iv. Ears forming low on the head.
- v. Growth abnormalities in the arms and legs.
- vi. Low muscle tone (hypotonia).
- vii. Small head and lower jaw.
- viii. Very small, close together or underdeveloped eyes.

Medical Management of Trisomy 13

1. Comprehensive Assessment and Diagnosis
2. Symptom-Specific Management
3. Palliative Care
4. Multidisciplinary Team Approach

Pediatricians, neonatologists, genetic counselors, cardiologists, neurologists, and palliative care specialists collaborate to provide comprehensive care.

Nursing Management of Trisomy 13

1. Assessment
2. Promoting Comfort
3. Nutritional Support
4. Preventing Complications
5. Family Support and Education
6. End-of-Life Care

TRISOMY 21

Trisomy 21 (Down syndrome) is a condition in which a person has an extra copy of chromosome 21. In most cases of Down syndrome, a child gets an extra chromosome 21 — for a total of 47 chromosomes instead of 46. Rarely, the extra chromosome 21 attaches to another chromosome. This extra genetic material causes the physical features and developmental delays in people with Down syndrome.

Clinical Features of Trisomy 21

Distinct physical signs of Down syndrome are usually present at birth and become more apparent as the baby grows. They can include facial features, such as:

- A flattened face, especially the bridge of the nose
- Almond-shaped eyes that slant up
- A tongue that tends to stick out of the mouth

Other physical signs can include:

A short neck, small ears, hands, and feet, a single line across the palm of the hand (palmar crease), small pinky fingers, poor muscle tone or loose joints, shorter-than-average height

Medical Management of Trisomy 21

1. Diagnosis

Prenatal:

- Screening tests (e.g., nuchal translucency, maternal serum screening).
- Diagnostic tests (e.g., amniocentesis, chorionic villus sampling).

Postnatal:

- Clinical features (hypotonia, upward slanting eyes, flat facial profile).
- Confirmation with karyotyping or genetic testing.

2. Addressing Common Medical Issues

- i. Cardiac Defects (40-50% of cases).
- ii. Gastrointestinal Issues:
- iii. Respiratory Issues:

3. Multidisciplinary Care

- Collaboration with pediatricians, cardiologists, neurologists, therapists, and educators for holistic care.

Nursing Management of Trisomy 21

1. Comprehensive Assessment
2. Promoting Growth and Development
3. Supporting Nutritional Needs
4. Preventing Infections
5. Monitoring for Complications

TRISOMY 18

Trisomy 18 (Edwards Syndrome) is a genetic disorder caused by the presence of an extra copy of chromosome 18. It is the second most common trisomy after Down syndrome and is associated with severe developmental delays, multiple congenital anomalies, and a high mortality rate. Many infants with Trisomy 18 do not survive beyond the first year of life. Management focuses on supportive and palliative care to ensure the comfort and quality of life of affected infants

Clinical Features of Trisomy 18

- i. Craniofacial anomalies: Microcephaly, low-set ears, cleft lip/palate, prominent occiput.
- ii. Limb abnormalities: Clenched fists with overlapping fingers, rocker-bottom feet.
- iii. Cardiac defects: Ventricular septal defect, patent ductus arteriosus, and other congenital heart diseases.
- iv. Neurological issues: Severe intellectual disability, hypotonia followed by hypertonia.
- v. Gastrointestinal issues: Omphalocele, esophageal atresia, and feeding difficulties.

Types of Trisomy 18

1. Full Trisomy 18 (95% of cases):
 - All cells have an extra copy of chromosome 18.
2. Mosaic Trisomy 18 (1-2% of cases):
 - Some cells have the extra chromosome, while others have a normal chromosomal makeup.
 - Symptoms are often milder.
3. Partial Trisomy 18 (rare):
 - A segment of chromosome 18 is duplicated and attached to another chromosome.

Nursing Management of Trisomy 18

1. Comprehensive Assessment:
2. Promoting Comfort and Safety:
3. Nutritional Support:
4. Preventing Infections:
5. Family Education and Support:
6. End-of-Life Care:

PHOCOMELIA

Phocomelia is a rare congenital condition characterized by the underdevelopment or absence of

the limbs. It typically results in shortened, malformed limbs that may resemble flippers. The term “phocomelia” originates from the Greek words phoke (seal) and melos (limb), reflecting the limb’s appearance.

Historical Note on Thalidomide

- The drug thalidomide was used in the late 1950s and early 1960s as a treatment for morning sickness during pregnancy. It was later discovered to cause severe birth defects, including phocomelia. This tragedy led to stricter drug regulations worldwide.

Causes of Phocomelia

1. Genetic Causes:
2. Environmental Causes:
3. Multifactorial Causes:

Types of Phocomelia

1. Complete Phocomelia: Absence of long bones in the limbs, with hands or feet attached directly to the trunk.
2. Partial Phocomelia: Some limb structures are present but are severely shortened or malformed.
3. Unilateral or Bilateral: Can affect one side (unilateral) or both sides (bilateral).

Associated Symptoms

1. Malformations of bones and joints in the arms or legs.
2. May occur with other congenital anomalies, such as cleft palate, heart defects, or spine abnormalities.
3. In some syndromic cases, developmental delays or other systemic issues may be present.

Diagnosis

- Prenatal Ultrasound: Can identify limb malformations early during pregnancy.
- Genetic Testing: Used to confirm syndromic forms or specific genetic mutations.

Management and Treatment

- Prosthetics and Orthotics.
- Physical Therapy.
- Surgical Interventions.

Nursing Management of Phocomelia

1. Assessment
 - Physical Assessment
 - Developmental Assessment
 - Psychosocial Assessment
 2. Promoting Mobility and Functionality
 3. Preventing Complications
 4. Nutritional Support
- THANK YOU FOR LISTENING

Paper 4 of 5 — NSG 503 LECTURER: MRS OYEKALE TOPIC: CNS ABNORMALITIES GROUP MEMBERS

Level 500 • First Semester • Handout

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COURSE CODE: NSG 503
LECTURER: MRS OYEKALE
TOPIC: CNS ABNORMALITIES
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INTRODUCTION

- Neural tube defects are the commonest malformations of the central nervous system. They arise from abnormalities during formation and closure of the neural tube.
- Ingestion of folic acid supplements prior to conception and during the early stages of pregnancy has helped to reduce the incidence of such anomalies.
- Prenatal screening is very effective at identifying these malformations and some parents choose to continue with their pregnancy while some choose selective termination of pregnancies where severe neural tube defects are found.

SPINA BIFIDA

- Spina bifida is a congenital neural tube defect (NTD) that occurs when the spine and spinal cord fail to form and close properly during early fetal development.
- It can lead to physical and neurological impairments of varying severity depending on the type and location of the defect.
- Normally, the neural tube closes completely by the 28th day of gestation.
- In spina bifida, incomplete closure occurs, leaving a portion of the spinal cord exposed or malformed.

TYPES OF SPINA BIFIDA

1. Spina Bifida Occulta: This is the mildest form which involves a small gap in the vertebrae without spinal cord or nerve damage. Often asymptomatic and discovered incidentally.
2. Meningocele: This is the moderate form of spina bifida. The meninges protrude through the spinal opening but do not involve the spinal cord. It usually has mild or no neurological deficits.
3. Myelomeningocele: This is the most severe form Spina bifida. Both the meninges and spinal cord protrude through the opening, often with nerve damage. It is associated with significant neurological and physical impairments.

RISK FACTORS

1. Low folic acid levels during pregnancy
2. Maternal obesity or diabetes
3. Family history of neural tube defects
4. Exposure to certain medications (e.g., valproic acid) Maternal hyperthermia (e.g., from fever or sauna use)

CLINICAL MANIFESTATIONS

● Spina Bifida Occulta:

1. Often asymptomatic
2. May present with a tuft of hair, dimple, or birthmark over the affected area.

● Meningocele:

1. Visible sac protrusion on the back.
2. Generally minimal neurological issues.

● Myelomeningocele:

1. Visible sac with exposed nerves and meninges.
2. Muscle weakness or paralysis below the defect.
3. Loss of bladder and bowel control.
4. Hydrocephalus (accumulation of cerebrospinal fluid in the brain).
5. Scoliosis or orthopedic deformities (e.g., clubfoot).

DIAGNOSIS

● Prenatal Diagnosis:

1. Elevated maternal alpha-fetoprotein (AFP) levels during pregnancy.
2. Ultrasound imaging to detect neural tube defects. Amniocentesis to confirm the diagnosis.

● Postnatal Diagnosis:

1. Physical examination for visible defects.
2. X-rays, MRI, or CT scans to assess the spinal cord and vertebrae.

COMPLICATIONS

1. Neurological Issues: Paralysis, sensory deficits, and learning disabilities.
2. Infections: Meningitis or urinary tract infections. Orthopedic Problems: Scoliosis, joint contractures, and fractures.
3. Psychosocial Impact: Emotional distress for the child and family

PREVENTION

1. Folic Acid Supplementation: Women of childbearing age should take 400–800 mcg of folic acid daily, starting before conception.
2. Healthy Lifestyle: Maintain a balanced diet and healthy weight. Avoid harmful substances like alcohol and tobacco.
3. Preconception Counseling: Identify and address risk factors in women planning pregnancy.

HYDROCEPHALUS

- Hydrocephalus is a medical condition characterized by an abnormal accumulation of cerebrospinal fluid (CSF) within the brain's ventricles, leading to increased intracranial pressure and potential damage to brain tissues.
- It can occur at any age but is most common in infants and older adults.
- This condition arises from a blockage in the circulation and absorption of cerebrospinal fluid, which is produced from the choroid plexuses within the lateral ventricles of the brain.
- The large lateral ventricles increase in size and eventually compress the surrounding brain tissue.
- Imbalance between CSF production and absorption leads to fluid buildup.
- This increases ventricular size and intracranial pressure, causing compression of brain tissues.

TYPES OF HYDROCEPHALUS

1. Communicating Hydrocephalus: CSF flows freely between ventricles but is not properly absorbed by the arachnoid villi. The common causes are meningitis, subarachnoid hemorrhage.
2. Non-communicating (Obstructive) Hydrocephalus: Blockage in CSF flow within the ventricular system. The common causes are brain tumors, congenital malformations (e.g., aqueductal stenosis).
3. Normal Pressure Hydrocephalus (NPH): Occurs in adults, characterized by normal intracranial pressure but with symptoms of hydrocephalus. Triad of symptoms are gait disturbance, urinary incontinence, and cognitive decline.
4. Congenital Hydrocephalus: Present at birth due to genetic or developmental abnormalities (e.g., neural tube defects, Chiari malformations).
5. Acquired Hydrocephalus: Develops later due to trauma, infections, tumors, or hemorrhage

RISK FACTORS

1. Premature birth (increased risk of intraventricular hemorrhage).
2. Infections during pregnancy (e.g., toxoplasmosis, cytomegalovirus).

CLINICAL MANIFESTATIONS

In Infants:

1. Enlarged head circumference.
2. Bulging or large tense anterior fontanelle.
3. Poor feeding, vomiting.
4. Irritability or lethargy.
5. "Sunsetting" eyes (downward deviation of the eyes).
6. Splayed skull sutures

In Normal Pressure Hydrocephalus:

1. Gait instability.
2. Urinary incontinence.
3. Dementia-like symptoms

DIAGNOSIS

● Imaging Studies:

1. Ultrasound: Used in infants with open fontanelles.
2. MRI/CT Scan: To detect ventricular enlargement and identify the underlying cause.

● Lumbar Puncture: Measures CSF pressure and can be therapeutic in NPH.

COMPLICATIONS

1. Shunt-related: blockage or malfunction, infection (meningitis or peritonitis).
2. Neurological: Persistent developmental delays or cognitive impairments, seizures.

PREVENTION

1. Prenatal Care: Adequate folic acid supplementation to reduce neural tube defects.
2. Avoid infections during pregnancy through vaccination and hygiene.
3. Early Detection: Routine ultrasound in high-risk pregnancies.

ANENCEPHALY

- Anencephaly is a severe congenital condition classified as a neural tube defect (NTD).
- It occurs when the upper part of the neural tube fails to close during early embryonic development, typically within the first 23–26 days of gestation.
- This defect results in the absence of a significant portion of the brain (the forebrain), vault of the skull and scalp.
- The neural tube is a structure in the developing embryo that eventually forms the brain and spinal cord.
- In anencephaly, the failure of neural tube closure affects the cranial end, leading to the incomplete development of the brain and skull.
- The forebrain and cerebrum, responsible for cognition, vision, and voluntary actions, are largely missing, leaving the brainstem exposed and functional to a limited extent.

RISK FACTORS

1. Inadequate folic acid intake during pregnancy
2. Genetic predispositions
3. Maternal exposure to teratogens (e.g., certain medications, alcohol)
4. Maternal diabetes
5. Obesity

CLINICAL MANIFESTATIONS

● Antenatal Diagnosis:

1. Detected via ultrasound as early as the first trimester
2. Absence of cranial bones and brain structures
3. Increased alpha-fetoprotein (AFP) levels in maternal blood or amniotic fluid

● Postnatal Appearance:

1. Absent scalp and cranial vault
2. Exposed brain tissue, often degenerated
3. Flat skull base

PREVENTION

1. Folic Acid Supplementation: Taking folic acid before conception and during early pregnancy reduces the risk of NTDs by up to 70%.
2. Preconception Counseling: Identify and mitigate risk factors such as diabetes or medication use.
3. Education: Public health campaigns about the importance of folic acid and prenatal care.

MICROCEPHALY

- Microcephaly is a medical condition in which an infant's head circumference is significantly smaller than the average for their age and sex.
- This condition often reflects an underdeveloped brain and may lead to developmental delays and other neurological complications.
- It can be present at birth (congenital) or develop postnatally due to slowed brain growth.
- In microcephaly, restricted brain growth limits skull expansion.

TYPES

1. Primary Microcephaly: Caused by genetic abnormalities affecting brain development during pregnancy.
2. Secondary Microcephaly: Occurs after birth due to brain injury or infection.

ETIOLOGY

Microcephaly results from various genetic, environmental, and infectious causes:

1. Genetic Factors: Chromosomal abnormalities (e.g., Down syndrome), Autosomal recessive genetic mutations
2. Infections During Pregnancy: Intrauterine infection, Zika virus (most notable recent cause), Cytomegalovirus (CMV), Rubella
- Maternal Health Issues: Severe malnutrition, Uncontrolled diabetes, substance abuse (e.g., alcohol, drugs)
3. Environmental Exposures: Radiation, Toxic chemicals
4. Other Causes: Perinatal asphyxia (oxygen deprivation), Severe jaundice, traumatic brain injury.

CLINICAL MANIFESTATIONS

1. Head Size: Head circumference significantly below average (measured using growth charts).
2. Neurological Signs: Developmental delays (motor and cognitive skills), Intellectual disability, Seizures, Hyperactivity, feeding difficulties.
3. Physical Features: Sloping forehead, Reduced brain mass evident on imaging (e.g., MRI)

DIAGNOSIS

- Prenatal Diagnosis:
 1. Ultrasound in the second or third trimester may detect microcephaly.

2. Amniocentesis for infections or genetic testing.

● Postnatal Diagnosis:

1. Measurement of head circumference at birth and during follow-up visits.
2. Imaging (MRI or CT) to assess brain structure.
3. Genetic and metabolic testing to identify underlying causes.

MIDWIVES' ROLE IN MANAGING CNS ABNORMALITIES

● ANTENATAL ROLE

1. Education and Counseling
2. Early Detection
3. Supportive Counseling
4. Preparation for Delivery

● INTRAPARTUM ROLE

1. Safe Delivery
2. Immediate Neonatal Care

● POSTNATAL ROLE

1. Initial Care
2. Parental Education
3. Psychosocial Support
4. Follow-Up and Developmental Monitoring

REFERENCES

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- Myles Textbook for Midwives 16th Edition

Paper 5 of 5 — ADVANCED MATERNAL AND CHILD HEALTH NURSING COURSE CODE: NSG 503

Level 500 • First Semester • Handout

OSUN STATE UNIVERSITY

FACULTY OF NURSING SCIENCE

DEPARTMENT OF MATERNAL AND CHILD HEALTH NURSING

ABNORMAL CONDITIONS OF THE NEW-BORN :

GASTROINTESTINAL DISORDERS

COURSE: ADVANCED MATERNAL AND CHILD HEALTH NURSING

COURSE CODE: NSG 503

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UMBILICAL HERNIA

• An umbilical hernia occurs when part of your intestine bulges through the opening in your abdominal muscles near your bellybutton (navel). An umbilical hernia creates a soft swelling or bulge near the navel. It occurs when part of the intestine protrudes through the umbilical opening in the abdominal muscles. Umbilical hernias in children are usually painless.

SYMPTOMS OF UMBILICAL HERNIA

- Visible Bulge
- Pain
- Reducibility

CAUSES OF UMBILICAL HERNIA

- Premature infants
- Low birthweight

Other causes in adults are; obesity, multiple pregnancies, fluid in abdominal cavity etc.

DIAGNOSIS

- Physical examination
- Ultrasound
- CT Scans/ MRI

TREATMENT

Many Umbilical Hernia closes on their own by age 3 to 4. If closure does not occur at the time,

surgical repair is advised.

CLEFT LIP

- Cleft lip, also known as cheiloschisis, is a congenital deformity that occurs when the tissues of the upper lip don't properly come together during fetal development. This results in a noticeable opening or split in the lip.

SYMPTOMS OF CLEFT LIP

1. A noticeable opening or split in the upper lip, which can be unilateral (one-sided) or bilateral (two-sided)
2. The opening can extend from the lip into the nose
3. Difficulty with feeding, speaking, or breathing in severe cases
4. Dental problems, such as missing or misaligned teeth
5. Hearing problems due to middle ear infections or other issues

CAUSES AND RISK FACTORS

- Genetics
- Environmental factors
- Nutritional deficiency
- Combination of genetics and environmental factors

DIAGNOSIS

- Physical Examination
- Ultrasound
- Imaging tests

TREATMENT

- Surgery
- Follow-up care
- Speech therapy and dental care

CLEFT PALATE

- Cleft palate is a congenital deformity that occurs when the tissues of the roof of the mouth (palate) don't properly come together during fetal development. This results in an opening or split in the palate, which can affect the upper jaw, lip, and nose.

SYMPTOMS

- An opening or split in the roof of the mouth (palate)
- Difficulty with feeding, swallowing, or speaking
- Nasal sounding speech due to air escaping through the opening
- Difficulty with hearing due to middle ear infections or other issues
- Dental problems, such as missing or misaligned teeth
- Difficulty with breathing in severe cases

CAUSES

- Genetics
- Environmental factors
- Nutritional deficiency

DIAGNOSIS

- Physical examination
- Ultrasound
- Imaging tests

TREATMENT

- Surgery
- Speech therapy
- Support and counseling

FRENULUM LINGUAE

• Frenulum linguae, also known as ankyloglossia or tongue-tie, is a congenital oral anomaly characterized by an unusually short or thick lingual frenulum. The lingual frenulum is a small fold of tissue that connects the underside of the tongue to the floor of the mouth.

SYMPTOMS

- Restricted tongue movements
- Speech difficulties
- Difficult with oral motor skills

CAUSES

- Genetic predisposition
- Congenital conditions, present at birth
- Abnormal development of the lingual frenulum during fetal development

DIAGNOSIS

- Physical examination by a healthcare provider, such as a pediatrician, dentist, or ENT specialist
- Observation of tongue movement and oral motor skills
- Assessment of speech and feeding abilities

TREATMENT

1. Frenectomy
2. Frenuloplasty
3. Laser treatment
4. Speech therapy
5. Oral motor therapy

DIAPHRAGMATIC HERNIA

• Diaphragmatic hernia, also known as congenital diaphragmatic hernia (CDH), is a birth defect that occurs when there is a hole in the diaphragm, the muscle that separates the chest cavity from the abdominal cavity. This hole allows organs from the abdomen, such as the stomach, small intestine, and liver, to move into the chest cavity, which can put pressure on the developing lungs.

SYMPTOMS

- Respiratory distress
- Difficulty feeding or vomiting

- Abdominal contents visible in the chest cavity on imaging tests
- Scaphoid abdomen (sunken abdomen)

CAUSES

- Genetic mutations or chromosomal abnormalities
- Environmental factors

DIAGNOSIS

- Prenatal ultrasound or MRI to detect the hernia before birth
- Physical examination after birth to assess respiratory status
- Chest X-ray or CT scan to confirm the diagnosis
- Echocardiogram to evaluate heart function

TREATMENT

- Surgical repair: closing the hole in the diaphragm and returning the abdominal organs to their normal position
- Stabilization of the infant: providing oxygen and respiratory support, as well as treating any related complication
- Extracorporeal membrane oxygenation (ECMO): using a machine to take over the work of the lungs and heart
- Long-term follow-up: monitoring for potential complications, such as respiratory problems or developmental delays

OBSTRUCTION OF THE ALIMENTARY TRACT

- Obstruction of the alimentary tract, also known as intestinal obstruction, is a condition where there is a blockage in the small or large intestine, preventing the normal flow of food, fluids, and gas.

SYMPTOMS

1. Severe abdominal pain or cramping
2. Nausea and vomiting
3. Abdominal distension or swelling
4. Constipation or inability to pass gas
5. Abdominal tenderness or guarding (muscle tension)

CAUSES

- Mechanical obstruction: physical blockage
- Functional obstruction: impaired intestinal motility

DIAGNOSIS

1. Physical examination and medical history
2. Imaging tests: X-rays, (CT) scan, (MRI), Ultrasound, Endoscopy or colonoscopy

TREATMENT

1. Fluid resuscitation and electrolyte replacement
2. Nasogastric suction to decompress the intestine
3. Medications to manage pain, nausea, and vomiting

4. Surgery to relieve the obstruction, such as:
5. Supportive care, such as:
 - Bowel rest
 - Parenteral nutrition (intravenous feeding)

ATRESIA OF THE ESOPHAGUS

- Atresia of the esophagus, also known as esophageal atresia (EA), is a rare congenital disease characterized by the abnormal development of the esophagus during fetal growth. This condition occurs when the esophagus does not fully form, resulting in a blind-ending upper esophageal pouch and a lower esophageal pouch that may or may not connect to the stomach.

SYMPTOMS

1. Polyhydramnios: Excessive amniotic fluid during pregnancy
2. Feeding difficulties: Choking, coughing, or regurgitation during feeding
3. Respiratory distress: Difficulty breathing, grunting, or retracting
4. Cyanosis : Bluish discoloration of the skin
5. Abdominal distension: Swelling of the abdomen

CAUSES

1. Genetic mutations
2. Environmental factors: Maternal nutrition, exposure to toxins, or infections during pregnancy

DIAGNOSIS

- Prenatal ultrasound
- Chest X-ray
- Esophagram
- Endoscopy
- Echocardiogram

TREATMENT

1. Surgical repair
2. Tracheoesophageal fistula repair
3. Gastrostomy tube placement
4. Supportive care

PYLORIC STENOSIS

- Pyloric stenosis is a congenital disease characterized by the thickening of the pylorus, the muscular valve that separates the stomach from the small intestine. This thickening causes narrowing of the pyloric channel, obstructing the passage of food into the intestine.

SYMPTOMS

1. Projectile vomiting: Forceful ejection of stomach contents
2. Regurgitation: Return of food into the mouth
3. Abdominal distension: Swelling of the abdomen

4. Dehydration: Excessive loss of fluids
5. Electrolyte imbalance: Disturbance in the balance of essential minerals
6. Weight loss: Failure to gain weight or weight loss
7. Hunger: Persistent hunger due to inability to retain food

CAUSES

1. Genetic predisposition: Family history of pyloric stenosis
2. Environmental factors
3. Hormonal influences: Imbalance of hormones that regulate muscle growth

DIAGNOSIS

1. Physical examination: Palpation of the abdomen to feel the thickened pylorus
2. Imaging tests: Ultrasound and barium swallow
3. Laboratory tests: Blood tests to evaluate electrolyte levels and renal function

TREATMENT

1. Surgical pyloromyotomy
2. Laparoscopic pyloromyotomy
3. Open pyloromyotomy

Postoperative care: Monitoring for complications and management of pain and nausea

Follow-up care: Regular check-ups to ensure proper healing and weight gain

OMPHALOCELE

- Omphalocele, also known as exomphalos, is a congenital disease characterized by a birth defect in which the intestines, liver, or other organs protrude through the navel (umbilicus) due to a gap in the abdominal wall.

SYMPTOMS

1. Visible bulge: Protrusion of intestines or organs through the navel
2. Abdominal contents exposed: Organs or intestines visible outside the abdominal wall
3. Increased risk of infection: Exposure of internal organs to the environment
4. Respiratory distress: Difficulty breathing due to compression of the lungs
5. Feeding difficulties: Inability to feed due to intestinal obstruction or other complications

CAUSES

1. Genetic mutations: Chromosomal abnormalities or genetic syndromes
2. Environmental factors: Exposure to certain substances or toxins during pregnancy
3. Combination of genetic and environmental factors

DIAGNOSIS

1. Prenatal ultrasound: Detection of omphalocele during fetal development
2. Physical examination: Visible bulge or exposed abdominal contents after birth
3. Imaging tests: X-rays, CT scans, or MRI to evaluate the extent of the defect
4. Chromosomal analysis: Genetic testing to identify potential chromosomal abnormalities

TREATMENT

1. Surgical repair

2. Staged repair: Multiple surgeries to gradually close the defect
3. Silastic silo placement:
4. Supportive care
5. Long-term follow-up

GASTROSCHISIS

- Gastroschisis is a congenital disease characterized by a birth defect in which the abdominal wall fails to close completely, resulting in a hole or defect, usually to the right of the belly button. This defect allows the intestines and sometimes other organs to protrude outside the abdominal cavity.

SYMPTOMS

1. Visible defect: A hole or defect in the abdominal wall, usually to the right of the belly button
2. Protruding intestines: Intestines and sometimes other organs, such as the stomach or liver, protrude outside the abdominal cavity
3. Increased risk of infection: Exposure of internal organs to the environment
4. Respiratory distress: Difficulty breathing due to compression of the lungs
5. Feeding difficulties: Inability to feed due to intestinal obstruction or other complications

CAUSES

1. Genetic mutations: Chromosomal abnormalities or genetic syndromes
2. Environmental factors: Exposure to certain substances or toxins during pregnancy
3. Combination of genetic and environmental factors

DIAGNOSIS

1. Prenatal ultrasound: Detection of gastroschisis during fetal development
2. Physical examination: Visible defect and protruding intestines after birth
3. Imaging tests: X-rays, CT scans, or MRI to evaluate the extent of the defect
4. Chromosomal analysis: Genetic testing to identify potential chromosomal abnormalities

TREATMENT

1. Surgical repair
2. Staged repair
3. Silastic silo placement
4. Supportive care
5. Long-term follow-up

THANKS FOR LISTENING